

Chapter 7

Rate of Return Analysis: Multiple Alternatives

This chapter presents methods to evaluate two or more alternatives using a rate of return (ROR) comparison. While a correctly performed ROR evaluation yields the same selection as Present Worth (PW) and Annual Worth (AW) analyses, its computational procedure differs significantly. Crucially, simply selecting the alternative with the highest individual ROR will not necessarily identify the best option. An alternative might have a high ROR but generate very little total value. Therefore, to compare alternatives on the basis of ROR, an incremental analysis is required.

7.1 Why Incremental Analysis Is Necessary

When comparing **mutually exclusive alternatives**, incremental analysis ensures proper comparison as highlighted in the following example.

Example 7.1. Sarah's father gives her a gift of \$10,000 and motivates her to put it in a savings account with a guaranteed APR of 3%. However, Sarah, having taken the Engineering Economy course, learned that she can do better, but with some risks. Hence, she is evaluating two investment alternatives:

A: Peer-to-peer lending opportunity that requires an investment of \$5,000 and has an internal rate of return i_A^* of 20% per year.

B: Investment in a stock portfolio that requires \$9,000 and has an i_B^* of 15% per year.

Intuitively, Sarah might conclude that the better alternative is the one with the larger

return, which is A in this case. However, this is not necessarily true.

While A has the higher projected return, its initial investment (\$5,000) is much less than her total available savings (\$10,000). What happens to the investment capital that is left over?

It is assumed that her excess funds will be invested at her baseline MARR of 3% (the savings account). Using this assumption, it is possible to determine the true financial consequences of the two alternative investments.

If alternative A is selected, \$5,000 will yield a return of 20% per year. The remaining \$5,000 will be invested at her MARR of 3% per year. The overall rate of return on her total savings will be the weighted average. Thus, if alternative A is selected, the overall ROR is calculated as follows:

$$\begin{aligned}\text{Overall ROR}_A &= \frac{5,000(0.20) + 5,000(0.03)}{10,000} \\ &= 11.50\%\end{aligned}$$

Conversely, if alternative B is selected, \$9,000 will be invested at 15% per year, and the remaining \$1,000 will earn 3% per year. The weighted average for alternative B is:

$$\begin{aligned}\text{Overall ROR}_B &= \frac{9,000(0.15) + 1,000(0.03)}{10,000} \\ &= 13.80\%\end{aligned}$$

This demonstrates that alternative B, despite having a lower individual rate of return, yields a higher overall return on Sarah's total available capital.

Note – Present Worth Verification

We can also verify this conclusion by calculating the Present Worth (PW) of each alternative over a one-year period. For alternative A, the initial \$5,000 investment grows by 20%:

$$\begin{aligned} PW_A &= -5,000 + \frac{5,000(1.20)}{1.03} \\ &= \$825.24 \end{aligned}$$

For alternative B, the initial \$9,000 investment grows by 15%:

$$\begin{aligned} PW_B &= -9,000 + \frac{9,000(1.15)}{1.03} \\ &= \$1,048.54 \end{aligned}$$

Since $PW_B > PW_A$, the PW analysis aligns perfectly with the overall ROR comparison.

Important – Individual Projects

When independent projects are evaluated, no incremental analysis is necessary between projects.

7.2 Calculation of Incremental Cash Flows

The incremental ROR method strictly requires that the alternatives be evaluated over an equal-service study period (such as the LCM of their useful lives). Hence, for unequal life alternatives, the cash flows must be developed using the LCM or the study period methods (see Section ?? of Ch. ??).

Definition – Incremental Cash Flow

The incremental cash flow is the difference between two cash flows, A and B , where alternative B has the higher initial investment:

$$\begin{aligned} \text{Incremental Cash Flow} &= \text{Cash Flow}_B - \text{Cash Flow}_A \\ \Delta CF_t &= CF_{B,t} - CF_{A,t} \quad \text{for } t = 0, 1, 2, \dots, \text{LCM} \end{aligned}$$

The incremental cash flow represents the extra investment or cost required if the alternative with the larger first cost is selected.

7.3 ROR Evaluation Using PW

7.3.1 Comparing Two Alternatives

Let Δi_{B-A}^* denote the ROR of the incremental cash flow series ΔCF_t .

Definition

If the rate of return available through the incremental cash flow equals or exceeds the MARR, the alternative associated with the extra investment should be selected:

- $\Delta i_{B-A}^* < MARR \rightarrow$ Select A.
- $\Delta i_{B-A}^* \geq MARR \rightarrow$ Select B.

The rationale for making the selection decision is the same as if only one alternative is under consideration, that alternative being the one represented by the incremental cash flow series. When viewed in this manner, it is obvious that unless this investment yields a rate of return equal to or greater than the MARR, the extra investment should not be made.

Note – Why Compare to MARR, Not Zero?

The MARR represents the **opportunity cost of capital**: it is the return your money would earn if invested elsewhere. For instance, U.S. Treasury bonds offer a virtually risk-free return and can serve as a natural MARR baseline. A positive return alone is not sufficient: the project must *outperform* this guaranteed alternative.

If $\Delta i_{B-A}^* = 5\%$ but $MARR = 10\%$, the extra capital invested in B earns less than it would in the baseline. You would be better off choosing the cheaper alternative and investing the difference at the MARR rate.

Not only must the return on the extra investment meet or exceed the $MARR$, but also the return on the investment that is common to both alternatives must meet or exceed the $MARR$. Accordingly, prior to performing an incremental ROR analysis, it is advisable to determine the internal rate of return i^* for each alternative.

Important – Revenue Alternatives Case

This can be done **only** for *revenue alternatives*, because cost alternatives have only cost (negative) cash flows and no positive i^* can be determined. For cost alternatives, skip this step and proceed directly to the incremental pairwise comparisons.

Example 7.2 (Incremental Cash Flow Analysis). A company is evaluating two mutually exclusive machines. The MARR is 10% per year. The cash flow details are as follows:

	Machine A	Machine B
First Cost	-\$10,000	-\$25,000
Annual Revenue	\$5,000	\$9,000
Salvage Value	\$2,000	\$5,000
Life (years)	5	5

Solution:

Since both alternatives are revenue alternatives, determine the individual ROR for each. For Machine A:

$$PW_A = -10,000 + 5,000(P/A, i, 5) + 2,000(P/F, i, 5) = 0 \Rightarrow i_A^* = 43.11\%$$

For Machine B:

$$PW_B = -25,000 + 9,000(P/A, i, 5) + 5,000(P/F, i, 5) = 0 \Rightarrow i_B^* = 26.54\%$$

Both i_A^* and i_B^* exceed the MARR of 10%, so both alternatives are economically viable. At this point, Machine A has the higher ROR, but the incremental analysis must be performed to determine which is the better choice.

Since Machine B has the higher first cost, the incremental cash flow is computed as $B - A$:

Year	Machine A	Machine B	Incremental (B-A)
0	-\$10,000	-\$25,000	-\$15,000
1	\$5,000	\$9,000	\$4,000
2	\$5,000	\$9,000	\$4,000
3	\$5,000	\$9,000	\$4,000
4	\$5,000	\$9,000	\$4,000
5	\$7,000	\$14,000	\$7,000

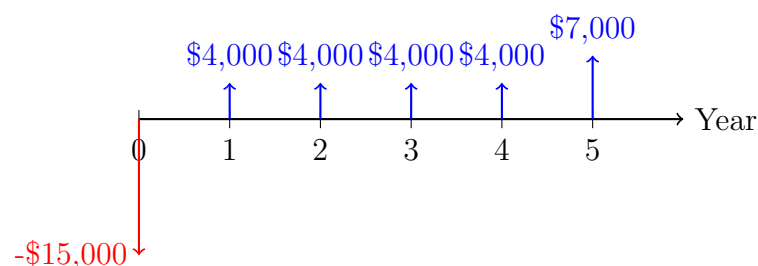


Figure 7.1: Cash Flow Diagram for the Incremental Series (B-A).

To find Δi_{B-A}^* , decompose the incremental series into a uniform \$4,000/yr over 5 years plus the \$3,000 salvage value difference at year 5, and solve:

$$PW_{\Delta} = -15,000 + 4,000(P/A, i, 5) + 3,000(P/F, i, 5) = 0$$

Initial Guess Procedure: To establish a starting bracket, convert the cash flows into a single equivalent annuity by ignoring the time value of money ($i = 0$):

$$\begin{aligned} A_{eq} &\approx 4,000 + \frac{3,000}{5} \\ &= \$4,600 \end{aligned}$$

Setting the initial cost equal to this equivalent annuity yields:

$$\begin{aligned} 15,000 &= 4,600(P/A, i, 5) \\ (P/A, i, 5) &= \frac{15,000}{4,600} \\ &= 3.261 \end{aligned}$$

Looking at the interest tables for $n = 5$, $(P/A, 16\%, 5) = 3.2743$. Thus, 16% is a great starting guess.

Trial at $i = 16\%$:

$$\begin{aligned} PW_{\Delta} &= -15,000 + 4,000(3.2743) + 3,000(0.4761) \\ &= -\$474.51 < 0 \end{aligned}$$

Since the PW is negative, the actual rate must be lower. Let us try $i = 14\%$:

$$\begin{aligned} PW_{\Delta} &= -15,000 + 4,000(3.4331) + 3,000(0.5194) \\ &= \$290.43 > 0 \end{aligned}$$

Interpolating between 14% and 16%:

$$\Delta i_{B-A}^* = 14\% + \frac{290.43}{290.43 + 474.51}(2\%) = 14.76\%$$

Since $\Delta i_{B-A}^* = 14.76\% \geq MARR = 10\%$, the extra investment in Machine B is justified. Therefore, **select Machine B**.

Note

Despite Machine A having a significantly higher individual ROR (43.11% vs. 26.54%), Machine B is the correct choice. The incremental analysis shows that the additional \$15,000 investment earns 14.76%, which exceeds the MARR.

Solution by Excel This problem is solved using the IRR function in Excel. Each alternative's net cash flows are entered in a column, and the incremental column is computed as $B - A$. The IRR function is applied to each column:

$$=IRR(B5:B10) \rightarrow 43.11\% \quad =IRR(C5:C10) \rightarrow 26.54\% \quad =IRR(D5:D10) \rightarrow 14.74\%$$

Incremental Cash Flow Analysis			
MARR = 10%			
Year	Machine A	Machine B	Incremental (B-A)
0	-10,000	-25,000	-15,000
1	5,000	9,000	4,000
2	5,000	9,000	4,000
3	5,000	9,000	4,000
4	5,000	9,000	4,000
5	7,000	14,000	7,000
Rate of Return Analysis			
	Machine A	Machine B	Incremental (B-A)
IRR (i*)	43.11%	26.54%	14.74%
Decision	≥ MARR ✓	≥ MARR ✓	≥ MARR → Select B

Figure 7.2: Excel solution for the incremental ROR analysis.

Example 7.3 (Incremental Analysis with Unequal Lives). Consider two revenue alternatives with different service lives. The MARR is 10% per year.

	Machine A	Machine B
First Cost	-\$8,000	-\$20,000
Annual Revenue	\$6,000	\$7,000
Salvage Value	\$1,000	\$3,000
Life (years)	3	6

Solution:

Since the lives differ, the LCM of 3 and 6 is **6 years**. Machine A must be purchased

twice over the 6-year study period: once at year 0 and again at year 3.

The individual ROR for each alternative (computed over its own life) is:

$$i_A^* = 57.72\% \quad i_B^* = 28.00\%$$

Both exceed the MARR, so both are viable candidates.

At year 3, Machine A reaches the end of its first life. The net cash flow at that point combines three items: the salvage of the old machine (\$1,000), the purchase of the replacement (-\$8,000), and the annual revenue (\$6,000). This yields a net of $6,000 + 1,000 - 8,000 = -\$1,000$.

The LCM cash flows and the incremental series are:

Year	Machine A	Machine B	Incremental (B-A)
0	-\$8,000	-\$20,000	-\$12,000
1	\$6,000	\$7,000	\$1,000
2	\$6,000	\$7,000	\$1,000
3	-\$1,000	\$7,000	\$8,000
4	\$6,000	\$7,000	\$1,000
5	\$6,000	\$7,000	\$1,000
6	\$7,000	\$10,000	\$3,000

The incremental series can be decomposed as a uniform \$1,000/yr for 6 years, plus \$7,000 at year 3, plus \$2,000 at year 6:

$$PW_{\Delta} = -12,000 + 1,000(P/A, i, 6) + 7,000(P/F, i, 3) + 2,000(P/F, i, 6) = 0$$

Initial Guess Procedure: Since calculating an equivalent annuity for this irregular cash flow series is prone to error (due to the large mid-period cash flow at year 3), it is safer to convert the cash flows into a single equivalent future sum (F) at year 6, ignoring the time value of money ($i = 0$):

$$\begin{aligned} F_{eq} &\approx (1,000 \times 6) + 7,000 + 2,000 \\ &= \$15,000 \end{aligned}$$

Setting the initial cost equal to this equivalent future sum yields:

$$12,000 = 15,000(P/F, i, 6)$$

$$\begin{aligned}(P/F, i, 6) &= \frac{12,000}{15,000} \\ &= 0.800\end{aligned}$$

Looking at the interest tables for $n = 6$, $(P/F, 4\%, 6) = 0.7903$. Thus, 4% is an excellent starting guess.

Trial at $i = 4\%$:

$$\begin{aligned}PW_{\Delta} &= -12,000 + 1,000(5.2421) + 7,000(0.8890) + 2,000(0.7903) \\ &= \$1,045.70 > 0\end{aligned}$$

Since the PW is highly positive, the true root is greater than 4%. Let us jump to test $i = 6\%$:

Trial at $i = 6\%$:

$$\begin{aligned}PW_{\Delta} &= -12,000 + 1,000(4.9173) + 7,000(0.8396) + 2,000(0.7050) \\ &= \$204.58 > 0\end{aligned}$$

Trial at $i = 8\%$:

$$\begin{aligned}PW_{\Delta} &= -12,000 + 1,000(4.6229) + 7,000(0.7938) + 2,000(0.6302) \\ &= -\$559.96 < 0\end{aligned}$$

Interpolating between 6% and 8%:

$$\Delta i_{B-A}^* = 6\% + \frac{204.58}{204.58 + 559.96}(2\%) = 6.54\%$$

Since $\Delta i_{B-A}^* = 6.54\% < MARR = 10\%$, the extra investment in Machine B is **not** justified. Therefore, **select Machine A**.

Note

Even though Machine B avoids the reinvestment hassle and provides steady cash flows, the incremental return of 6.54% does not meet the MARR. The cheaper Machine A, replaced every 3 years, is the economically superior choice.

Visualizing the Decision (PW vs. MARR): Plotting the Present Worth of both alternatives across various MARR values provides a clear visual explanation of the incremental decision (see Figure 7.3). The blue curve represents the PW of Machine A and

the red curve represents that of Machine B.

- The curves intersect at exactly $MARR = 6.54\%$, which is the incremental rate of return (Δi_{B-A}^*). This is the “breakeven” interest rate.
- For any $MARR < 6.54\%$, the PW of Machine B (red curve) is higher than Machine A, meaning the extra investment in B is economically justified.
- For any $MARR > 6.54\%$ (including the company’s actual MARR of 10%), the PW of Machine A (blue curve) is higher, confirming that Machine A is the superior choice.

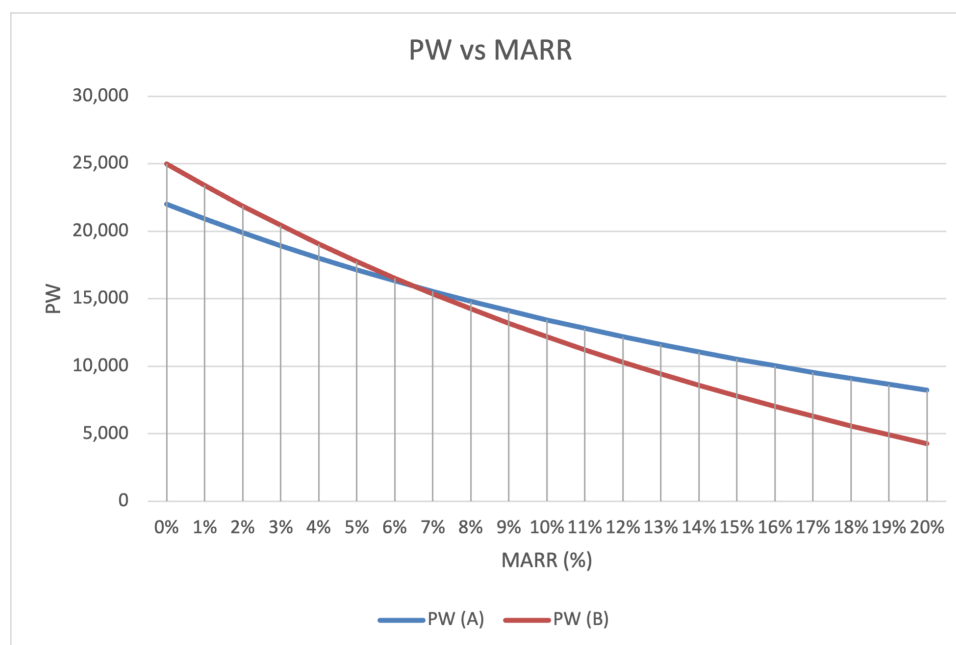


Figure 7.3: Present Worth of Machine A and Machine B plotted against the MARR. The crossover point at 6.54% is the incremental rate of return.

Note – Steps for Incremental ROR Analysis

The procedure for an incremental ROR analysis of two mutually exclusive alternatives is as follows:

1. Order the alternatives by increasing first cost.
2. **(Revenue alternatives only)** Compute i^* for each alternative. Any alternative with $i^* < MARR$ is eliminated.
3. Calculate the incremental cash flow: $\Delta F_t = CF_{B,t} - CF_{A,t}$, where B has the higher first cost.
4. Determine the incremental rate of return Δi_{B-A}^* from the incremental cash flow series.
5. If $\Delta i_{B-A}^* \geq MARR$, select the higher-cost alternative (B). Otherwise, select the lower-cost alternative (A).

7.3.2 Comparing Multiple Alternatives

When more than two mutually exclusive alternatives exist, the incremental ROR method extends naturally by performing successive pairwise comparisons. The alternatives are ranked by initial investment, and the winning alternative from each comparison advances to face the next challenger.

The steps for incremental ROR evaluation for multiple alternatives are summarized below:

1. Rank the alternatives from **smallest to largest** initial investment.
2. **(Revenue alternatives only)** Compute i^* for each alternative. Discard any alternative with $i^* < MARR$.
3. Compute the incremental cash flow between the first surviving alternative and the next one (using the LCM of their lives if they differ).
4. Determine Δi^* . If $\Delta i^* \geq MARR$, the challenger wins; otherwise, the current alternative survives.
5. Repeat steps 3–4 with the winner against the next alternative until all have been compared.

Important – The “Do Nothing” Baseline

For revenue alternatives, the “Do Nothing” (DN) alternative is implicitly the first defender with an initial cost of \$0. Screening out alternatives with an individual $i^* < MARR$ is mathematically equivalent to performing an incremental comparison against DN. If all individual ROR values fall below the MARR, all alternatives lose to DN, and no investment should be made.

Example 7.4 (Multiple Alternatives with Different Lives). A logistics company is considering four mutually exclusive Automated Storage and Retrieval Systems to upgrade its main warehouse. The MARR is 10% per year.

Alternative	Initial Cost	Annual Revenue	Salvage	Life
B (Semi-Automated)	-\$2,500,000	\$1,100,000	\$200,000	3 yr
D (Refurbished)	-\$3,500,000	\$950,000	\$150,000	4 yr
C (Standard AS/RS)	-\$5,000,000	\$1,600,000	\$400,000	4 yr
A (Fully Automated)	-\$7,000,000	\$1,900,000	\$500,000	6 yr

The alternatives are already ordered by increasing initial cost.

Solution:

Step 1: Screen individual ROR values. Since these are revenue alternatives, compute i^* for each:

$i_B^* = 18.33\% \geq MARR$	viable
$i_D^* = 4.90\% < MARR$	discard
$i_C^* = 13.04\% \geq MARR$	viable
$i_A^* = 17.08\% \geq MARR$	viable

Alternative D is eliminated. The surviving alternatives are B, C, and A.

Step 2: Pairwise comparison C vs. B. The LCM of the remaining lives (3, 4, 6) is **12 years**. To find the 12-year incremental ROR, one could set up the PW of each alternative over 12 years and equate them, or use the IRR function in Excel on the expanded 12-year incremental cash flows ($C - B$). Solving $PW_C(i) - PW_B(i) = 0$ yields:

$$\Delta i_{C-B}^* = 7.26\%$$

Since $7.26\% < MARR$, the extra investment in C is not justified. **B survives.**

Step 3: Pairwise comparison A vs. B. Similarly, equating the PW of alternatives A and B over the 12-year LCM, $PW_A(i) - PW_B(i) = 0$, yields:

$$\Delta i_{A-B}^* = 16.37\%$$

Since $16.37\% \geq MARR$, the extra investment in A is justified. **A wins.**

Conclusion: Select Alternative A (Fully Automated system).

Note

Note that Alternative B had the highest individual ROR (18.33%), yet Alternative A (17.08%) is the correct choice. This illustrates exactly why incremental analysis is necessary: the alternative with the highest ROR is not always the best investment.

7.4 ROR Evaluation Using AW

The incremental ROR analysis can also be performed using the Annual Worth (AW) method instead of Present Worth. The primary advantage of using AW is that it naturally satisfies the equal-service requirement without needing the Lowest Common Multiple (LCM) of lives. Because the AW of one life cycle is equivalent to the AW over any number of identical life cycles, alternatives with different service lives can be compared directly using their single-cycle cash flows.

To find the incremental rate of return Δi_{B-A}^* using the AW method, set the Annual Worth of both alternatives equal to each other and solve for i :

$$\begin{aligned} AW_A(i) &= AW_B(i) \\ \Delta AW_{B-A}(i) &= AW_B(i) - AW_A(i) = 0 \end{aligned}$$

The resulting Δi^* will be exactly the same as the one found using the PW method over the LCM.

Note – Why Calculate ROR Instead of Just AW?

Since evaluating the AW (or PW) at the $MARR$ immediately identifies the best alternative and is computationally simpler, you might wonder why we perform ROR analysis at all. There are two primary reasons:

- **Management Preference:** Decision-makers often prefer percentage rates over absolute dollar amounts because percentages are easier to intuitively compare against corporate hurdle rates.
- **Sensitivity Analysis:** The incremental rate of return Δi^* is the exact “breakeven” interest rate between two alternatives. Knowing this value tells you exactly how sensitive your decision is to fluctuations in the $MARR$.

Example 7.5 (Incremental ROR using AW). Reconsider the pairwise comparisons from the previous AS/RS example. Instead of using the 12-year LCM, we can find the incremental ROR using the AW of their single life cycles.

Comparison C vs. B: Set $AW_C = AW_B$:

$$\begin{aligned} AW_C(i) &= -5,000,000(A/P, i, 4) + 1,600,000 + 400,000(A/F, i, 4) \\ AW_B(i) &= -2,500,000(A/P, i, 3) + 1,100,000 + 200,000(A/F, i, 3) \end{aligned}$$

Solving $AW_C(i) - AW_B(i) = 0$ yields:

$$\Delta i_{C-B}^* = 7.26\%$$

This exactly matches the PW result. Since $7.26\% < MARR$, B survives.

Comparison A vs. B: Set $AW_A = AW_B$:

$$\begin{aligned} AW_A(i) &= -7,000,000(A/P, i, 6) + 1,900,000 + 500,000(A/F, i, 6) \\ AW_B(i) &= -2,500,000(A/P, i, 3) + 1,100,000 + 200,000(A/F, i, 3) \end{aligned}$$

Solving $AW_A(i) - AW_B(i) = 0$ yields:

$$\Delta i_{A-B}^* = 16.37\%$$

Since $16.37\% \geq MARR$, Alternative A is the winner, perfectly mirroring the PW approach without expanding the cash flows to 12 years.

7.5 Summary

Evaluating mutually exclusive alternatives using the Rate of Return method requires a strict **incremental analysis**. Simply selecting the alternative with the highest individual rate of return (i^*) can lead to mathematically incorrect decisions, as it ignores the return generated by the uninvested capital (the MARR).

The key takeaways and procedures for incremental ROR analysis are:

- **Ranking and Baseline:** Alternatives must be ranked by increasing initial investment. For revenue alternatives, any option with an individual $i^* < MARR$ is eliminated, which is mathematically equivalent to selecting the “Do Nothing” baseline.
- **Pairwise Comparisons:** The incremental cash flow is defined as the cash flow of the higher-cost alternative minus the lower-cost alternative. If the incremental rate of return (Δi^*) equals or exceeds the MARR, the extra investment is justified, and the higher-cost alternative wins.
- **Unequal Lives:** When using Present Worth (PW) for the incremental evaluation, alternatives with unequal lives must be evaluated over the Lowest Common Multiple (LCM) of their lives.
- **The AW Shortcut:** Using the Annual Worth (AW) method to find the incremental ROR naturally satisfies the equal-service requirement. By setting $AW_A(i) = AW_B(i)$ over their respective single-cycle lives, the need for an LCM is completely bypassed, while yielding the exact same decision as the PW method.